

Supplementary Information

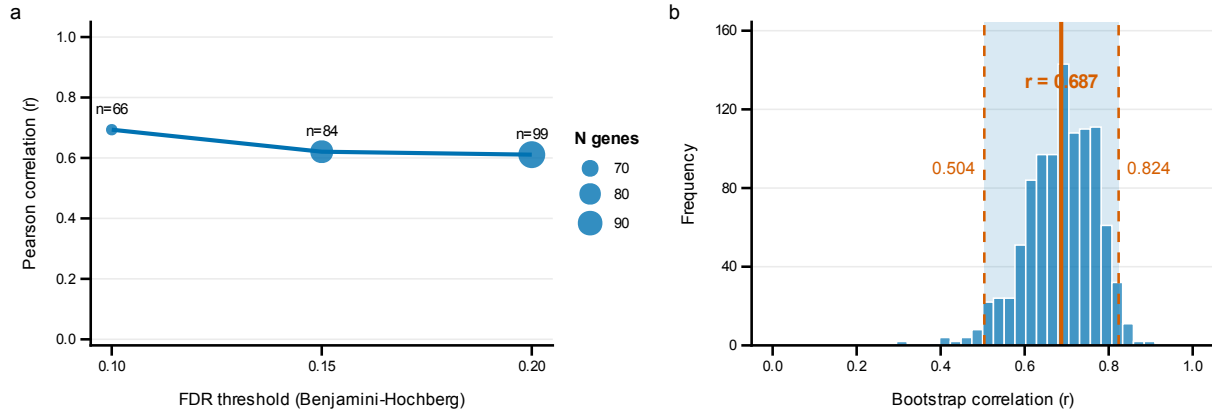
A Multi-Tissue Transcriptomic Atlas of Porcine Development Identifies Conserved Molecular Programs with Humans

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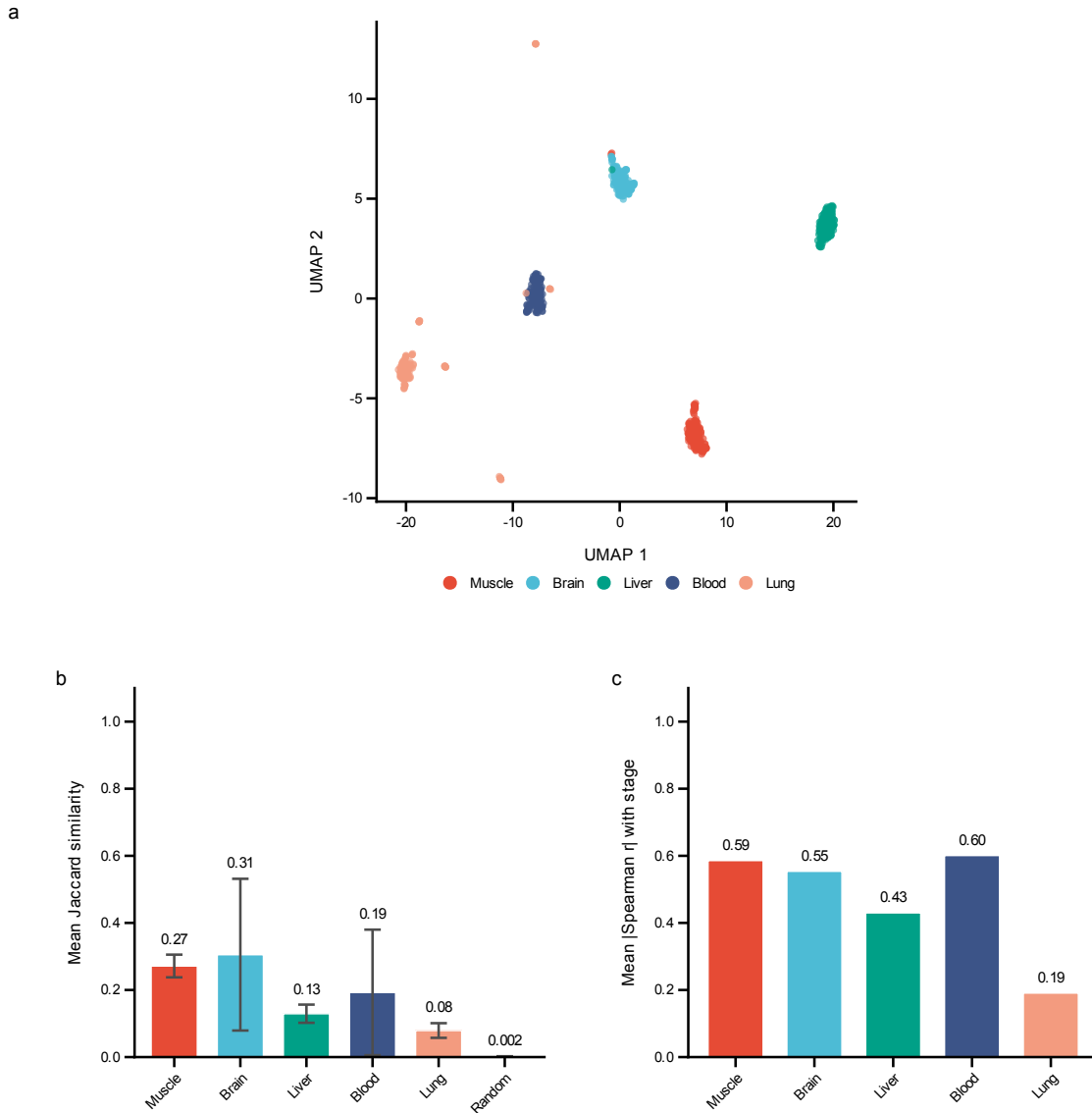
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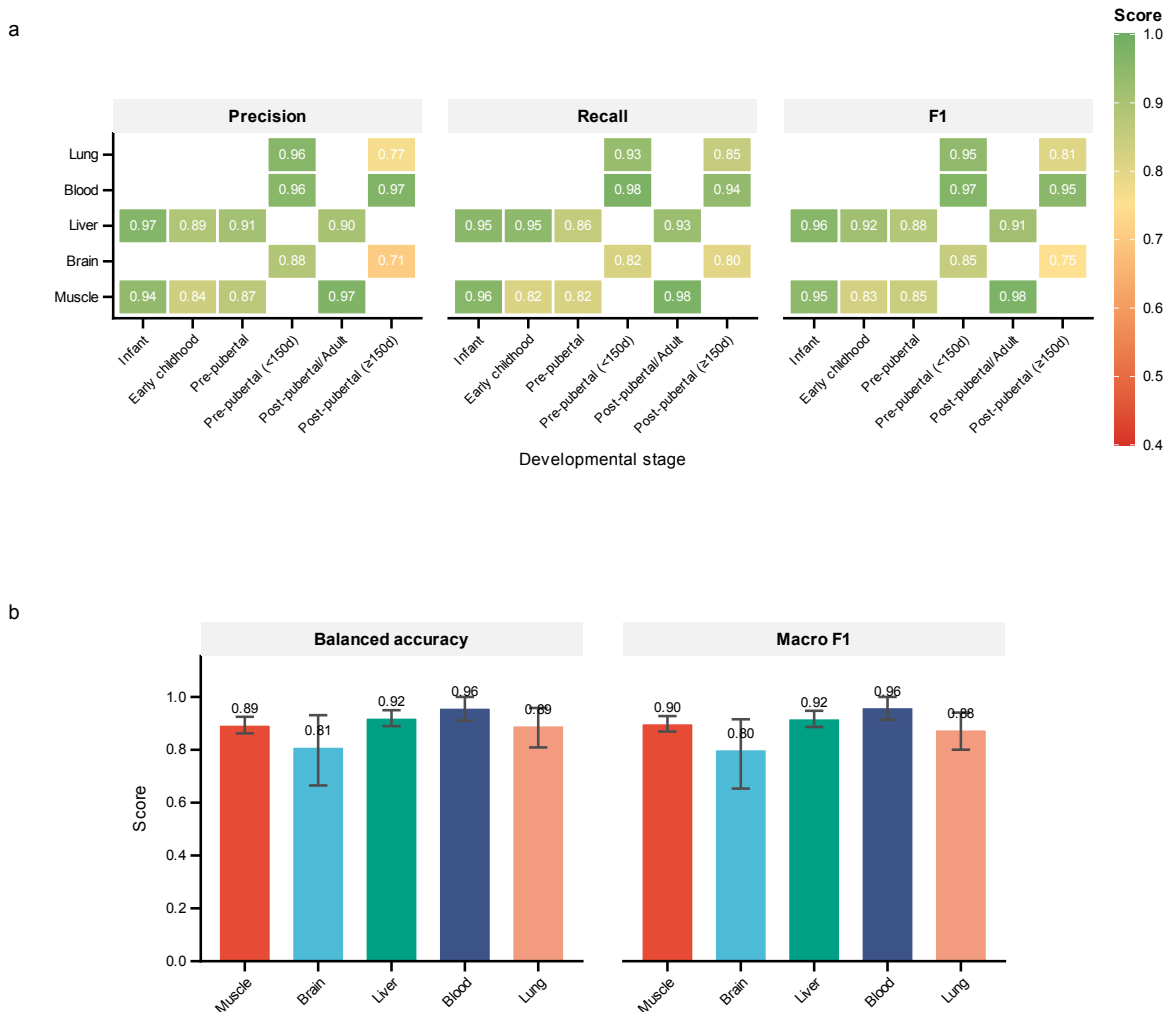
Supplementary Figures



Supplementary Figure 1: Cross-species sensitivity analysis of conserved developmental markers. a, Pearson correlation between pig and human developmental fold changes across FDR thresholds ranging from < 0.05 to < 0.20 (Benjamini–Hochberg corrected), using all genes passing pre-specified criteria (FDR $<$ threshold and $|\log_2 \text{FC}| > 0.5$ in both species). Points are sized by the number of genes at each threshold. **b**, Bootstrap distribution of correlation coefficients ($n = 1,000$ iterations) for the pre-specified criteria (FDR < 0.10 , $|\log_2 \text{FC}| > 0.5$, $n = 66$ genes). Solid vertical line indicates the observed correlation ($r = 0.693$); dashed lines denote the 95% bootstrap confidence interval boundaries (0.50–0.82).



Supplementary Figure 2: Tissue specificity and feature selection stability analysis. **a**, Uniform manifold approximation and projection (UMAP) of gene expression profiles across five tissues (muscle, brain, liver, blood, lung), demonstrating distinct tissue-specific clustering patterns ($n = 150$ samples per tissue, balanced sampling; top 500 variable genes). For visualization only, BioProject batch effects were removed per tissue using `limma::removeBatchEffect` prior to UMAP; the machine learning pipeline operates on uncorrected expression data. **b**, Within-tissue feature selection stability quantified as mean Jaccard similarity index of the top 50 selected features across ten independent random seeds. Error bars represent standard deviation; higher values indicate more reproducible feature selection. **c**, Mean absolute Spearman correlation between top-ranked features and developmental stage for each tissue.



Supplementary Figure 3: Extended classification performance metrics across tissues and developmental stages. **a**, Heatmap of per-class precision, recall, and F1 scores for each developmental stage across all five tissues. Colour scale ranges from 0.4 (red, poor performance) to 1.0 (green, excellent performance). Missing values indicate developmental stages not present in that tissue's classification scheme. **b**, Overall performance summary showing balanced accuracy and macro-averaged F1 scores evaluated on pooled cross-validated held-out predictions for each tissue. Error bars indicate 95% bootstrap confidence intervals from 1,000 resampling iterations.

Supplementary Tables

Supplementary Table 1: Train versus test performance comparison. Both train and test (pooled cross-validated held-out predictions) performance are reported for all tissues. The gap between train and test metrics indicates the degree of overfitting; regularisation and early stopping were used to minimise this gap.

Tissue	Set	Scheme	n	Balanced Acc.	F1-macro	F1-weighted
Muscle	Train	4-class	761	0.968	0.974	0.984
Muscle	Test	4-class	761	0.894	0.899	0.939
Brain	Train	2-class	43	0.915	0.922	0.930
Brain	Test	2-class	43	0.811	0.801	0.816
Liver	Train	4-class	309	0.985	0.987	0.987
Liver	Test	4-class	309	0.921	0.918	0.919
Blood	Train	2-class	78	1.000	1.000	1.000
Blood	Test	2-class	78	0.959	0.960	0.961
Lung	Train	2-class	129	0.948	0.942	0.961
Lung	Test	2-class	129	0.892	0.876	0.916

Supplementary Table 2: Per-class performance metrics by tissue. Precision, recall, and F1 scores for each developmental stage within each tissue. n indicates the total number of samples per class (pooled across cross-validation folds).

Tissue	Stage	n	Precision	Recall	F1
Muscle	infant	194	0.939	0.959	0.949
Muscle	early childhood	76	0.838	0.816	0.827
Muscle	pre-pubertal	90	0.871	0.822	0.846
Muscle	post-pubertal/adult	401	0.973	0.980	0.976
Brain	pre-pubertal (<150d)	28	0.885	0.821	0.852
Brain	post-pubertal ($\geq$ 150d)	15	0.706	0.800	0.750
Liver	infant	87	0.965	0.954	0.960
Liver	early childhood	58	0.887	0.948	0.917
Liver	pre-pubertal	83	0.910	0.855	0.882
Liver	post-pubertal/adult	81	0.904	0.926	0.915
Blood	pre-pubertal (<150d)	45	0.957	0.978	0.967
Blood	post-pubertal ($\geq$ 150d)	33	0.969	0.939	0.954
Lung	pre-pubertal (<150d)	102	0.960	0.931	0.945
Lung	post-pubertal ($\geq$ 150d)	27	0.767	0.852	0.807

Supplementary Table 3: Functional annotations of cross-species developmental markers. Annotations compiled from UniProt, GeneCards, and literature review. FC = log₂ fold change (adult/infant); positive values indicate higher expression in adults, negative values indicate higher expression in infants. Pig FC from pig GTEx data (GEO: GSE257558); Human FC from Schaiter *et al.* (2024). Genes ordered by model importance within each functional module. “—” indicates the gene did not pass cross-species criteria (FDR < 0.10, |log₂ FC| > 0.5) but is included based on its known biological role.

Gene	Developmental Function	Pig FC	Human FC
<i>Module 1: Neuromuscular Maturation and Growth Signalling</i>			
<i>ACHE</i>	Functions as the key enzyme at the neuromuscular junction and as a structural adhesion molecule guiding neuronal growth during development ¹ .	−3.34	−0.53
<i>NRK</i>	Embryonic Ste20-type kinase that promotes actin polymerisation via cofilin phosphorylation and activates JNK signalling during skeletal muscle formation; expression restricted to embryonic and early postnatal development ² .	−8.56	−2.45
<i>KREMEN1</i>	Negative regulator of Wnt/ β -catenin signalling, partnering with Dkk1 to control cell survival and ensure proper patterning during postnatal development ^{3,4} .	−1.69	−1.43
<i>IGF2BP2</i>	RNA-binding protein that promotes myoblast proliferation via translational control of growth-related transcripts (c-Myc, Igf1r, Sp1); postnatal downregulation required for terminal myogenic differentiation ⁵ .	−2.19	−2.03
<i>SORCS2</i>	VPS10 domain-containing receptor that binds progranulin to regulate motor neuron development during neuromuscular maturation ⁶ .	−4.67	−0.95
<i>ETV5</i>	FGF-responsive PEA3-subfamily ETS transcription factor that promotes stem/progenitor cell self-renewal and myogenic differentiation in cooperation with MEF2; highly expressed during active postnatal myogenesis ⁷ .	−2.80	−2.24
<i>IGF2BP3</i>	Oncofetal RNA-binding protein of the same family as IGF2BP2; expressed during embryonic development and silenced postnatally, consistent with its role in fetal growth programs.	−4.85	−1.23
<i>CCDC8</i>	Core component of the 3M complex (with CUL7 and OBSL1) that regulates microtubule dynamics, genome integrity, and growth hormone signalling during rapid postnatal growth; consistent with severe growth restriction in 3-M syndrome ⁸ .	−4.44	−3.14
<i>Module 2: Structural Refinement and ECM Remodelling</i>			
<i>TCAF1</i>	Regulates TRPM8 ion channel trafficking and Ca ²⁺ signalling; may support calcium-dependent processes during early postnatal muscle growth and metabolic maturation ⁹ .	−4.72	−1.02
<i>FGFRL1</i>	Required for slow muscle fibre development; loss of function leads to diaphragm defects due to impaired slow fibre specification ¹⁰ .	−3.03	−2.10
<i>WDR1</i>	Cofactor of cofilin-mediated actin filament disassembly, essential for sarcomere assembly and myofibril organisation during postnatal muscle growth; higher expression in immature muscle reflects intensive cytoskeletal remodelling during myofibrillogenesis ¹¹ .	−1.69	−0.52
<i>DLK1</i>	Crucial regulator of the myogenic program that inhibits myoblast proliferation while promoting differentiation; high infant expression declines as the muscle fiber population matures ^{12,13} .	−9.15	−2.88
<i>FSCN1</i>	Actin-bundling protein essential for formation of cellular protrusions during cell migration and tissue morphogenesis; highly expressed in embryonic somite-derived tissues, with postnatal downregulation as migratory programs complete ¹⁴ .	−4.28	−1.95
<i>POSTN</i>	Secreted matricellular protein that mediates ECM remodelling, collagen fibrillogenesis, and connective tissue maturation during postnatal myofiber development; expression declines as tissue architecture stabilises ¹⁵ .	−7.97	−2.93
<i>FREM2</i>	Basement membrane extracellular matrix protein of the FRAS/FREM complex maintaining epithelial–mesenchymal adhesion during organogenesis; postnatal downregulation reflects transition to collagen VII-dominated mature basement membranes ¹⁶ .	−4.58	−1.05
<i>ADAMTS2</i>	Procollagen N-proteinase that cleaves propeptides from fibrillar procollagens to initiate collagen fibrillogenesis; high infant expression reflects intensive ECM biosynthesis during rapid postnatal growth ¹⁷ .	−4.25	−1.83
<i>COL11A2</i>	Component of type XI collagen regulating collagen fibril organisation in cartilage and developing connective tissue; high infant expression reflects active skeletal morphogenesis and ECM assembly ¹⁸ .	−8.35	−0.70
<i>TRIM54</i>	Muscle-specific E3 ubiquitin ligase (MuRF3) involved in sarcomere Z-disc assembly; expression increases postnatally as the mature contractile apparatus is established.	—	—
<i>Module 3: Metabolic Transition</i>			
<i>ME1</i>	Malic enzyme 1, catalysing oxidative decarboxylation of malate to pyruvate; a key source of NADPH for lipid biosynthesis. Expression increases during development, reflecting metabolic maturation.	2.59	0.64
<i>MAOB</i>	Mitochondrial outer membrane enzyme catalysing oxidative deamination of monoamines; postnatal upregulation reflects developmental maturation of oxidative metabolism and the glycolytic-to-oxidative fiber-type shift.	2.55	0.80
<i>PLA2G4E</i>	Calcium-dependent N-acyltransferase synthesising N-acylphosphatidylethanolamines (endocannabinoid precursors); species-discordant expression may reflect divergent intramuscular lipid metabolism programs.	2.32	−1.25

Supplementary Table 4: Stage-by-stage expression trajectories of cross-species developmental markers. Mean TPM values across porcine developmental stages for genes with conserved pig-human developmental patterns. Stage boundaries: infant (0–20 d), early childhood (21–59 d), pre-pubertal (60–149 d), post-pubertal (150–365 d), adult (>365 d). “Peak Change” indicates the developmental transition with the largest fold change. Genes are grouped by functional module and ordered by model importance within each module.

Gene	infant	early	pre-pub	post-pub	adult	Peak Change
<i>Module 1: Neuromuscular Maturation and Growth Signalling</i>						
<i>ACHE</i>	17.5	7.6	3.4	1.3	1.7	infant→early
<i>NRK</i>	128.2	4.2	8.1	1.3	0.3	infant→early
<i>KREMEN1</i>	30.3	13.2	9.4	5.2	9.4	infant→early
<i>IGF2BP2</i>	11.3	1.5	1.9	1.0	2.5	infant→early
<i>SORCS2</i>	14.4	3.0	2.2	0.6	0.6	infant→early
<i>ETV5</i>	34.2	16.7	7.5	5.5	4.9	early→pre-pub
<i>IGF2BP3</i>	21.8	1.7	2.1	0.4	0.7	infant→early
<i>CCDC8</i>	177.4	28.2	25.4	13.4	8.2	infant→early
<i>Module 2: Structural Refinement and ECM Remodelling</i>						
<i>TCAF1</i>	78.9	8.3	8.1	3.8	3.0	infant→early
<i>FGFRL1</i>	77.4	15.6	14.2	4.3	9.5	infant→early
<i>WDR1</i>	202.9	125.6	95.0	51.9	63.1	pre-pub→post-pub
<i>DLK1</i>	338.8	3.9	6.4	0.7	0.6	infant→early
<i>FSCN1</i>	123.7	32.5	18.9	6.2	6.4	infant→early
<i>POSTN</i>	602.4	44.1	54.3	2.5	2.4	pre-pub→post-pub
<i>FREM2</i>	33.1	4.4	5.5	1.5	1.4	infant→early
<i>ADAMTS2</i>	51.1	16.6	16.0	6.4	2.7	infant→early
<i>COL11A2</i>	44.6	1.5	1.0	0.3	0.1	infant→early
<i>TRIM54</i>	78.3	351.1	149.4	377.9	144.4	infant→early
<i>Module 3: Metabolic Transition</i>						
<i>ME1</i>	2.8	4.0	3.9	6.9	16.9	post-pub→adult
<i>MAOB</i>	5.7	24.2	10.8	15.7	33.4	infant→early
<i>PLA2G4E</i>	0.1	1.4	0.7	0.6	0.5	infant→early

Supplementary Table 5: Within-project fold-change validation across developmental contrasts. Developmental fold-changes ($\log_2$ scale) were computed within single-project muscle cohorts—each spanning four developmental stages under uniform laboratory protocols—and correlated with full-dataset estimates across three stage comparisons (all using infant as baseline). Pooled within-project fold-changes were computed by averaging across both projects before correlating with full-dataset estimates. “All genes” includes all 31,908 expressed genes; “Cross-species” includes the 66 genes passing pre-specified criteria ($\text{FDR} < 0.10$, $|\log_2 \text{FC}| > 0.5$) in both pig and human. CI = 95% bootstrap confidence interval (1,000 iterations). Per-project results are available in the analysis repository. Full-dataset sample sizes: infant $n = 194$, early childhood $n = 76$, pre-pubertal $n = 90$, post-pubertal $n = 396$ (excludes 5 adult samples merged into post-pubertal/adult in the 4-class scheme).

Comparison	Gene set	Full dataset		Within-project		Pooled r [95% CI]	p -value
		n_{Inf}	n_{later}	n_{Inf}	n_{later}		
infant vs. early childhood	All genes	194	76	106	12	0.669 [0.66–0.68]	$< 10^{-300}$
	Cross-species					0.761	1.3×10^{-13}
infant vs. pre-pubertal	All genes	194	90	106	27	0.806 [0.80–0.82]	$< 10^{-300}$
	Cross-species					0.965	6.9×10^{-39}
infant vs. post-pubertal	All genes	194	396	106	10	0.799 [0.79–0.81]	$< 10^{-300}$
	Cross-species					0.978	5.6×10^{-45}

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